MongoDB Capstone Project 2

Project Title

Business Scenario

An EdTech platform wants to analyze:

```
Learners
Courses
Enrollments
Payments
Course Progress
Quiz Scores
Instructor Performance
Completion Rates
Revenue
```

The goal is to build MongoDB collections and perform analytics using find queries, array queries, updates, aggregation, and `$lookup`.

Database Name

```
use edtech_capstone_db
```

Required Collections

```
learners
courses
```

```
enrollments
instructors
```

Collection 1: learners

```
db.learners.insertMany([
  {
    learner_id: 1,
    name: "Rahul Sharma",
    city: "Hyderabad",
    experience_years: 2,
    goal: "Data Engineer",
    phone: "9876543210"
  },
  {
    learner_id: 2,
    name: "Priya Reddy",
    city: "Bangalore",
    experience_years: 4,
    goal: "AI Engineer",
    phone: "9876543211"
  },
  {
    learner_id: 3,
    name: "Amit Kumar",
    city: "Mumbai",
    experience_years: 1,
    goal: "Data Analyst",
    phone: null
  },
  {
    learner_id: 4,
    name: "Sneha Patel",
    city: "Chennai",
    experience_years: 6,
    goal: "ML Engineer",
    phone: "9876543213"
  },
  {
    learner_id: 5,
    name: "Farhan Ali",
```

```
    city: "Delhi",
    experience_years: 3,
    goal: "Cloud Engineer",
    phone: "9876543214"
  },
  {
    learner_id: 6,
    name: "Meera Nair",
    city: "Pune",
    experience_years: 0,
    goal: "AI Engineer",
    phone: null
  }
])
```

Collection 2: instructors

```
db.instructors.insertMany([
  {
    instructor_id: 101,
    instructor_name: "Abdullah Khan",
    expertise: ["AI", "Data Engineering", "Cloud"],
    rating: 4.9
  },
  {
    instructor_id: 102,
    instructor_name: "Neha Singh",
    expertise: ["Power BI", "SQL", "Analytics"],
    rating: 4.6
  },
  {
    instructor_id: 103,
    instructor_name: "Ravi Kumar",
    expertise: ["Python", "Machine Learning"],
    rating: 4.7
  }
])
```

Collection 3: courses

```
db.courses.insertMany([
  {
    course_id: 201,
    course_name: "Data Engineering with Azure",
    category: "Data Engineering",
    instructor_id: 101,
    price: 15000,
    level: "Intermediate",
    tools: ["SQL", "Python", "Azure Data Factory", "Databricks"]
  },
  {
    course_id: 202,
    course_name: "AI Engineer Roadmap",
    category: "Artificial Intelligence",
    instructor_id: 101,
    price: 20000,
    level: "Beginner",
    tools: ["Python", "OpenAI", "Vector DB", "LangChain"]
  },
  {
    course_id: 203,
    course_name: "Power BI for Business",
    category: "Analytics",
    instructor_id: 102,
    price: 8000,
    level: "Beginner",
    tools: ["Power BI", "Excel", "SQL"]
  },
  {
    course_id: 204,
    course_name: "Machine Learning Practical",
    category: "Machine Learning",
    instructor_id: 103,
    price: 12000,
    level: "Intermediate",
    tools: ["Python", "Scikit-learn", "Pandas"]
  },
  {
    course_id: 205,
    course_name: "Cloud AI Engineer",
```

```
    category: "Cloud",
    instructor_id: 101,
    price: 18000,
    level: "Advanced",
    tools: ["Azure", "AWS", "GCP", "AI Services"]
  }
])
```

Collection 4: enrollments

```
db.enrollments.insertMany([
  {
    enrollment_id: 1001,
    learner_id: 1,
    course_id: 201,
    enrollment_date: ISODate("2026-01-10"),
    payment: {
      amount: 15000,
      mode: "UPI",
      status: "Success"
    },
    progress: {
      completed_modules: 8,
      total_modules: 10,
      completion_percent: 80
    },
    quiz_scores: [75, 82, 88],
    status: "Active"
  },
  {
    enrollment_id: 1002,
    learner_id: 2,
    course_id: 202,
    enrollment_date: ISODate("2026-01-15"),
    payment: {
      amount: 20000,
      mode: "Card",
      status: "Success"
    },
    progress: {
      completed_modules: 10,
```

```
    total_modules: 10,
    completion_percent: 100
  },
  quiz_scores: [90, 92, 95],
  status: "Completed"
},
{
  enrollment_id: 1003,
  learner_id: 3,
  course_id: 203,
  enrollment_date: ISODate("2026-02-01"),
  payment: {
    amount: 8000,
    mode: "Cash",
    status: "Pending"
  },
  progress: {
    completed_modules: 3,
    total_modules: 8,
    completion_percent: 37.5
  },
  quiz_scores: [60, 65],
  status: "Active"
},
{
  enrollment_id: 1004,
  learner_id: 4,
  course_id: 204,
  enrollment_date: ISODate("2026-02-10"),
  payment: {
    amount: 12000,
    mode: "UPI",
    status: "Success"
  },
  progress: {
    completed_modules: 6,
    total_modules: 12,
    completion_percent: 50
  },
  quiz_scores: [78, 80, 85],
  status: "Active"
},
{
  enrollment_id: 1005,
```

```
    learner_id: 5,
    course_id: 205,
    enrollment_date: ISODate("2026-03-05"),
    payment: {
      amount: 18000,
      mode: "Card",
      status: "Failed"
    },
    progress: {
      completed_modules: 0,
      total_modules: 12,
      completion_percent: 0
    },
    quiz_scores: [],
    status: "Payment Failed"
  },
  {
    enrollment_id: 1006,
    learner_id: 6,
    course_id: 202,
    enrollment_date: ISODate("2026-03-12"),
    payment: {
      amount: 20000,
      mode: "UPI",
      status: "Success"
    },
    progress: {
      completed_modules: 2,
      total_modules: 10,
      completion_percent: 20
    },
    quiz_scores: [55],
    status: "Active"
  }
])
```

Capstone Tasks

Part 1: Basic Find Queries

1. Display all learners.
2. Display all courses.
3. Display learner name, city, and goal only.
4. Find learners from Hyderabad.
5. Find learners whose goal is `AI Engineer`.
6. Find courses in the `Data Engineering` category.
7. Find courses priced above ₹10,000.
8. Find beginner-level courses.
9. Find enrollments with successful payments.
10. Find learners where phone is null.

Part 2: Operators

11. Find learners having experience greater than 2 years.
12. Find courses priced between ₹8,000 and ₹18,000.
13. Find courses where level is either `Beginner` or `Intermediate`.
14. Find enrollments where completion percent is greater than or equal to 80.
15. Find enrollments where payment status is not `Success`.
16. Find learners from Hyderabad, Bangalore, or Pune.
17. Find courses not in the `Cloud` category.

Part 3: Array Queries

18. Find instructors having expertise in `AI`.
19. Find instructors having expertise in `SQL`.
20. Find courses using tool `Python`.
21. Find courses using tool `Databricks`.
22. Find enrollments where quiz score contains `95`.
23. Find enrollments where any quiz score is greater than 85.

Part 4: Sorting and Limit

24. Sort courses by price descending.
25. Display top 3 most expensive courses.
26. Sort learners by experience years descending.
27. Display top 2 most experienced learners.

28. Sort instructors by rating descending.

Part 5: Update Operations

29. Update learner 1 city to `Secunderabad`.
30. Update course 203 price to `9000`.
31. Update enrollment 1006 completion percent to `30`.
32. Change enrollment 1005 status to `Inactive`.
33. Add field `active: true` to all learners.
34. Remove field `active` from all learners.
35. Add tool `MongoDB` to course 201.

Part 6: Delete Operations

36. Delete enrollments where payment status is `Failed`.
37. Delete learners whose experience years is 0.

Part 7: Count and Distinct

38. Count total learners.
39. Count total courses.
40. Count successful enrollments.
41. Display distinct learner cities.
42. Display distinct course categories.
43. Display distinct payment modes.

Part 8: Aggregation

44. Revenue by payment mode.

45. Revenue by course.

46. Count learners by goal.

47. Average course price by category.

48. Average completion percentage by course.

49. Count enrollments by status.

50. Courses having revenue greater than ₹15,000.

Part 9: Lookup / Join Style Queries

Use `$lookup`.

51. Enrollments with Learner Details

Display:

```
Enrollment ID
Learner Name
City
Course ID
Status
```

52. Enrollments with Course Details

Display:

```
Enrollment ID
Course Name
Category
Amount
Payment Status
```

53. Courses with Instructor Details

Display:

```
Course Name
Category
```

```
Instructor Name
Instructor Rating
```

54. Full Enrollment Report

Display:

```
Enrollment ID
Learner Name
City
Goal
Course Name
Category
Instructor Name
Payment Amount
Payment Status
Completion %
Enrollment Status
```
